

Re: harmonics on the three-sphere — how the “honest” matrices were built

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This note only addresses the $SO(3) / l = 1$ harmonic sign-transpose question raised in your PDF. It is a warm-up calculation on S^2 , isolating the operator/generator convention issue before we move to the full S^3 spinor-harmonic construction — it does not yet say anything about that larger question.

1 The two comparisons, side by side

Your own check computes, for a candidate abstract matrix T'_i and the combined column of basis functions $v = (\psi_{1,1}; \psi_{1,0}; \psi_{1,-1})$, the ordinary matrix–vector product $T'_i \cdot v$, and compares it entrywise to $(\hat{L}_i \psi_{1,1}; \hat{L}_i \psi_{1,0}; \hat{L}_i \psi_{1,-1})$.

That is a well-defined thing to compute, but it is a *different* operation from asking: “what matrix, in the basis $\{\psi_{1,1}, \psi_{1,0}, \psi_{1,-1}\}$, actually represents $\hat{L}_i$?”

The second question is answered like this: apply $\hat{L}_i$ to *one basis function at a time*, and expand the result back in the same basis:

$$\hat{L}_i \psi_{1,m} = \sum_{m'} M_{m'm} \psi_{1,m'},$$

so that column m of the matrix M is exactly the (unique) set of coefficients needed to write $\hat{L}_i \psi_{1,m}$ back in the basis. This M is the honest matrix representation of $\hat{L}_i$.

If $(T'_i \cdot v)_{\text{row}}$ happens to equal $\hat{L}_i \psi_{1,\text{row}}$ for every row, that tells you T'_i and M are related by a **transpose**: $(T'_i \cdot v)_{\text{row}} = \sum_{\text{col}} (T'_i)_{\text{row,col}} \psi_{1,\text{col}}$ uses the *rows* of T'_i , whereas the honest construction of M above reads off the *columns*. For a diagonal or otherwise symmetric matrix the row and column pictures coincide, which is exactly why your $\hat{L}_z = T'_z$ check gave a clean match — $T'_z = \text{diag}(1, 0, -1)$ is trivially symmetric. It is not a coincidence that the one clean component was the diagonal one.

2 Building M three separate computational routes

I built the honest matrix M_x (and M_y, M_z) three separate ways, to make sure the conclusion did not depend on which technique was used — these are three computational cross-checks rather than three fully independent derivations, since all three use the same operators, basis functions, and conventions throughout.

Method A — direct solve. Apply $\hat{L}_x$ to each of $\psi_{1,1} = \sin \theta e^{i\phi}$, $\psi_{1,0} = \cos \theta$, $\psi_{1,-1} = \sin \theta e^{-i\phi}$ in turn, then solve the linear system for the coefficients that re-express each result in the same basis (matching the $e^{i\phi}$, constant, and $e^{-i\phi}$ parts separately).

Method B — coefficient extraction. Same idea, but the coefficients are extracted by evaluating the residual at a few sample values of ϕ and solving directly, rather than a general symbolic solve — a cheap cross-check on Method A.

Method C — inner-product (bra-ket) projection. The cleanest of the three, and the one I would recommend as the standard going forward:

$$M_{m'm} = \frac{\langle \psi_{1,m'} | \hat{L}_i | \psi_{1,m} \rangle}{\langle \psi_{1,m'} | \psi_{1,m'} \rangle}, \quad \langle f | g \rangle := \int_0^\pi \int_0^{2\pi} \bar{f} g \sin \theta \, d\phi \, d\theta.$$

This is exactly the ordinary quantum-mechanical way of reading off a matrix element, and it sidesteps the row/column ambiguity entirely, since each entry of M is defined by its own independent integral. The simple form above (dividing by $\langle \psi_{1,m'} | \psi_{1,m'} \rangle$ alone) relies on $\{\psi_{1,1}, \psi_{1,0}, \psi_{1,-1}\}$ being an orthogonal set, which they are here; for a non-orthogonal basis one would need the full Gram-matrix inversion, $M = G^{-1}A$ with $A_{m'm} = \langle \psi_{m'} | \hat{L}_i | \psi_m \rangle$ and $G_{m'm} = \langle \psi_{m'} | \psi_m \rangle$.

All three methods gave, for your specific choice $\chi_{1,1} = \chi_{1,-1} = \sin \theta$, $\chi_{1,0} = \cos \theta$, exactly:

$$M_x = \begin{pmatrix} 0 & -\frac{1}{2} & 0 \\ -1 & 0 & 1 \\ 0 & \frac{1}{2} & 0 \end{pmatrix}, \quad M_y = \begin{pmatrix} 0 & \frac{i}{2} & 0 \\ -i & 0 & -i \\ 0 & \frac{i}{2} & 0 \end{pmatrix}, \quad M_z = \text{diag}(1, 0, -1).$$

Comparing these directly to your own T'_x, T'_y, T'_z (equations 19–21 of your note):

$$M_x = -(T'_x)^\top, \quad M_y = +(T'_y)^\top, \quad M_z = T'_z,$$

each an exact identity (zero residual), independent of which of the three methods above produced M .

3 Why y and z looked clean and x did not — the real reason

A clarification on terminology first: M above is the matrix acting on *coefficient* columns (if $\Psi = (\psi_{1,1}, \psi_{1,0}, \psi_{1,-1})$ is treated as the basis itself and a state is a coefficient vector c in that basis, M sends $c \mapsto Mc$). Your own comparison instead treats Ψ as the column being acted on directly, so it tests $M^\top$, not M — consistent with Section 1.

I initially described the match for y, z as “coincidental,” which was sloppy and not quite right. Since $\psi_{1,1}, \psi_{1,0}, \psi_{1,-1}$ are linearly independent, an exact match across all three rows is a genuine matrix identity, not a numerical accident tied to one test vector. The real explanation is more structural, and traces back to your own S matrix directly.

Identify $\psi_{1,1} = x + iy$, $\psi_{1,0} = z$, $\psi_{1,-1} = x - iy$ (exactly, with $x = \sin \theta \cos \phi$, $y = \sin \theta \sin \phi$, $z = \cos \theta$ — no extra normalization needed). Applying $\hat{L}_i$ to the plain Cartesian triple (x, y, z) gives, for *all three* generators at once:

$$\hat{L}_i(x, y, z)^\top = -T_i(x, y, z)^\top,$$

using your *original* (un-diagonalized) generators T_x, T_y, T_z from equations (14)–(16). This is a single, uniform sign — the standard fact that a differential operator realizing a rotation of *functions* carries the opposite sign to the same rotation acting on the coordinates themselves. It is not itself the source of the asymmetry between x and y, z .

The asymmetry comes from matching your S -diagonalized ordering to the $(\psi_{1,1}, \psi_{1,0}, \psi_{1,-1})$ ordering you actually compare against. Writing $X = (x, y, z)^\top$, direct computation gives $SX = (-\psi_{1,-1}, \psi_{1,0}, -\psi_{1,1})^\top$ — the *same* three functions as $\Psi = (\psi_{1,1}, \psi_{1,0}, \psi_{1,-1})^\top$, but reordered and with two sign flips. That reordering is exactly the matrix

$$C = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \quad C^2 = I, \quad \Psi = CSX.$$

Combining the uniform sign above with this reordering gives, for every generator at once,

$$\hat{L}_i \Psi = -C T'_i C^{-1} \Psi,$$

which I have checked directly matches $\hat{L}_i \Psi$ exactly for $i = x, y, z$. The point is what C -conjugation does to each T'_i :

$$C T'_x C^{-1} = T'_x \quad (\text{unchanged}), \quad C T'_y C^{-1} = -T'_y, \quad C T'_z C^{-1} = -T'_z.$$

Combined with the uniform “ -1 ” from the active/passive sign, x picks up $(-1)(+1) = -1$ relative to T'_x , while y, z pick up $(-1)(-1) = +1$ relative to T'_y, T'_z — exactly reproducing $M_x = -(T'_x)^\top$, $M_y = +(T'_y)^\top$, $M_z = T'_z$. The x -generator is singled out because C (forced by the specific ordering and sign convention you used for $\psi_{1,\pm 1}$) happens to commute with T'_x but anti-commute with T'_y and T'_z — a systematic consequence of the basis reordering, not an accident of specific numerical entries.

In short: your arithmetic in equations (1)–(29) is correct throughout — I rechecked every one of them independently and found no error — and the -1 on x alone has this precise, traceable origin, not an arbitrary one.

4 One small, unrelated thing, for completeness

Separately: in your equation (18), S^{-1} has a sign slip in one entry (the $(2, 3)$ position). It did not affect anything downstream — your own T'_x, T'_y, T'_z in equations (19)–(21) are exactly what you get from the *correct* inverse, so nothing needs to be revisited there.

5 Looking forward to your reasoning

I’d be very interested to see why you were expecting $\hat{L}_i \psi = T'_i \psi$ to hold identically for all three generators — if there’s a different, perhaps more physically natural convention where that holds exactly (rather than up to the transpose above), that would be worth understanding properly before we go further with the three-sphere case.